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Roll No

EX-601 (GS)
B.E. VI Semester Examination, June 2020
Grading System (GS)
Communication Engineering
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. The probability density function for a random variable X is given by $f(x) = x/10$ for $x = 1, 2, 3, 4$. Write out the probability distribution of X as a table and calculate the probability that X is less than 3, $P(X < 3)$.
2. Describe the working of the single sideband modulation with the help of necessary equations and sketches.
3. Define modulation index in AM. Draw and explain the modulated signals in case of:
 - i) $m_a < 1$
 - ii) $m_a = 1$
 - iii) $m_a > 1$
4. Find the Fourier transform of the following:
 - i) $x(2t)$
 - ii) $x(t - 2)e^{it}$
5. a) What are the drawbacks of TRF receiver? Draw block diagram and explain superheterodyning receivers.
b) Draw block diagram of PCM and explain each block. What is quantization error?
6. Explain :
 - i) Energy and power signals
 - ii) Deterministic and random signals
7. For a PCM system with following parameters, determine:
 - i) Minimum sample rate
 - ii) Minimum no. of bits used in the PCM code.Maximum analog input frequency = 4kHz.
Maximum decoded voltage at receiver = $\pm 2.55V$.
Minimum dynamic range = 46dB.
8. Write short notes on (any three):
 - a) Satellite Link calculation
 - b) VSB modulation
 - c) LASER
 - d) AGC and AVC
 - e) Parametric amplifiers

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